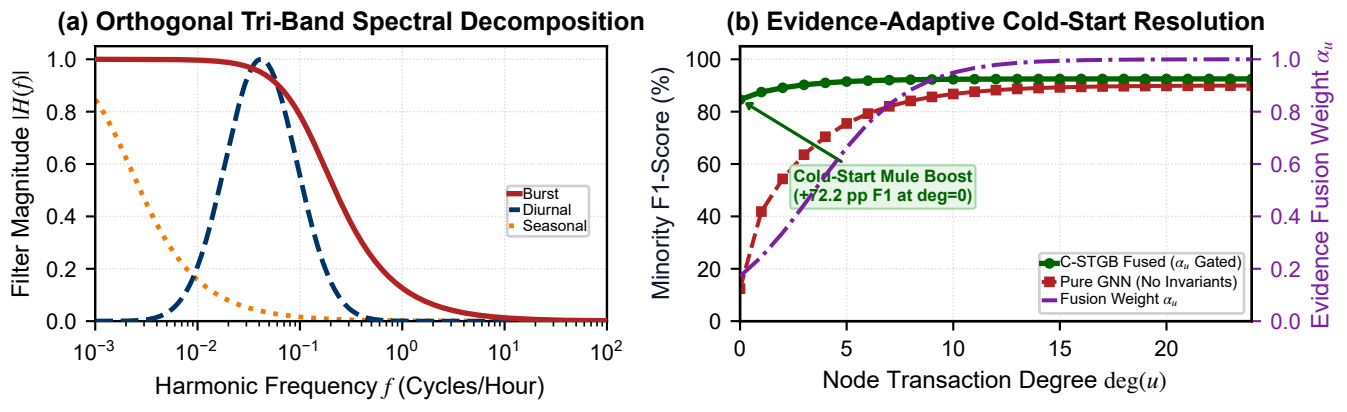

(a) Orthogonal Tri-Band Spectral Decomposition
Filter Magnitude |H(f)|
1.0
0.8
0.6
0.4
0.2
0.0
Burst
Diurnal
Seasonal
10^-3
10^-2
10^-1
10^0
10^1
10^2
Harmonic Frequency f (Cycles/Hour)
(b) Evidence-Adaptive Cold-Start Resolution
Minority F1-Score (%)
100
80
60
40
20
0
Cold-Start Mule Boost
(+72.2 pp F1 at deg=0)
C-STGB Fused (α_u Gated)
Pure GNN (No Invariants)
Fusion Weight α_u
0
5
10
15
20
Node Transaction Degree deg(u)
Evidence Fusion Weight α_u
1.0
0.8
0.6
0.4
0.2
0.0